

Final Report Requirements

Upload all files and documents to your team's Google drive submission folder.

Purpose

The purpose of this report is to provide a written record of your team's work on designing a solution for tracking the number of people or team-defined objectives. It is our intent to have future teams learn from prior semesters' efforts and project ideas, therefore you must thoroughly document and explain your ideas, why they worked/didn't work, how they worked, etc.

The length of your report is not critical. But it should contain sufficient detail for a *skilled electrical engineer* to be able to reproduce your results as well as to further develop your solution concept. For example, you should provide the detailed model number of your sensor instead of simply saying the generic name, such as a time-of-flight sensor. You should also briefly summarize the most important parameters of your sensor so that your reader does not need to comb through the entire datasheet by him/herself. In case the sensor becomes discontinued, your reader will be able to quickly identify a replacement with the necessary characteristics. To improve the readability, the report requires additional content and supporting files in the Appendices and/or as file attachments (submitted together with the report to the Google drive).

You are expected to provide extended discussions on why some test scenarios did not (always) work during your final demonstration, if applicable and especially when your instructor has explicitly made the request. The discussions should be based on scientific analysis or/and data rather than mere hypotheses. You should also provide possible solutions to improve your device's performance.

When explaining your algorithm, start with a high-level description of how it works using layman's terms before describing the technical details. It will significantly improve the readability of your report. The same principle applies to other parts of your report too.

Deadline

The final report is due on the 4/27 Monday midnight.

Content

We expect the following sections to appear in your final paper

Introduction

- Brief summary of the problems your design is trying to address, the goals it is trying to achieve, and the motivation behind them. (This is for documentation purposes and to make your report more complete and interesting to those you share it with, e.g. when applying for an internship or permanent position.)
- Summarize the technologies used in your final design.

Design Selection

- Discuss your general strategy for going from problem/area to possible solution to implemented solution. If your team used multiple strategies, discuss each. These strategies should not be specific to each idea, but rather your team's approach to solving problems as a whole.
- Give us detailed notes as to why each candidate approach was selected or eliminated from the running. Did you do an experiment? Did you analyze by calculating impact using some model? Both? Show the results and conclusion. Show your calculations. Was it a project management / team capabilities driven decision? Explain that.
- A note on strategies; good strategies are usually "tradeoffs". I.e, a specific strategy almost always comes at the expense of something else. Platitudes such as "we decided to work hard" or "we communicated well" are not meaningful, as all teams should strive to follow these best engineering principles. A good example of a strategy is "our team decided to focus on creating a systematic testing process for our designs at the beginning of the semester at the expense of actually getting started on final project design right away"

Final Prototype

- Provide a complete system block diagram and associated schematics for the final prototype you built for the final project demonstration
- Do not include systems which you did not have working and integrated by the project demonstration day (you are supposed to describe those in 'Design Selection' section)
- Explain how the entire system works together, including how/when different parts communicate, power consumption (peak, average), how environmental constraints were accounted for, etc.
- Discuss algorithms as appropriate here but document actual code in source files.

- Discuss the power management strategy such as any power-saving methods used. Include all analyses and the projection of the battery lifetime.
- Discuss how you or a future team could implement or continue to improve on this idea. Provide specific strategies, resources, etc.
- Provide the bill of materials (BoM) analysis of your final system which needs to be at most \$100 per unit. A penalty will be applied if BoM is higher. Note that BoM is different from what you actually spent to build your system. BoM is the total cost of materials (ESP32 boards, PCB, sensors, etc.) when you mass-produce your system.

Conclusion

- Discuss any high-level conclusions your team has drawn about the given problem, either of your solutions or the technology you used. For example, working on tracking people for a pandemic might have brought up some concerns about privacy, which your team had discussed at length and done some additional research into. Discuss thoughts such as these that have arisen from your work.
- Reflect on what your team did well and what you would want to do better in the future (Note that we all often learn more from “failure” than success. Show us what you learned). Specifically, try and identify strategies your team used that resulted in the most successful systems, good presentations, etc. Similarly, try and identify specific habits or actions your team took which hindered the achievement of your goals.
- What did you learn about the design process?
- What is one piece of advice you would give students in a future semester regarding this final project?

Team Work

- Specify the role and contributions of each team member

Appendices

You must include all schematics, PCB files, code, etc. These files must adhere to the following specifications and be uploaded into a subfolder named "Final paper team X supporting files". You must **also** make a list of everything you attached in the Appendix section of your paper (This does not count towards your text limit).

- All schematics must be clean, and submitted in both PDF and tool-specific digital file formats.
- Mechanical parts, if made, must be submitted in both CAD and 4 standard views PDFs

- All code must be provided in a single archive (i.e. all code should be in its own sub-folder within the submission zip); teams MUST TEST:
 - that opening the archive into a development environment allows the project to be compiled.
 - that the archive contains NO temporary and/or auto-generated files -- it should be the minimal set of usable source files.
- Source files must be clearly documented
- An overview README file in txt or PDF must be provided, listing all the files delivered and describing their use.

Specifications

- Length: about 8 pages excluding appendices ([This length is recommended but not enforced.](#)) single spaced. Note: Images, graphics, plots, etc. count towards your length.
- You may include a title page with a graphic / photo and a table of contents (ideally hyper-linked to the sections). This does not count towards the number of pages.
- Recommended format: 11pt Arial font, single spaced, 1" margins. Headers with 14pt font.
- **Please upload all files to the Google drive folder assigned to your team:**
 - Your report as a Word document, or a Google doc, or a PDF file. Name your file "Final paper team X".
 - All supporting source files, schematics, etc. in a subfolder named "Final paper team X supporting files".
- Equations, plots, figures, and graphics should all be labeled (*equation 1*, *plot 1*, etc.) and referred to as such in the text. Please pay close attention to equation typesetting quality.

Grading rubric

Category	Points
Introduction	15
Design Selection	25

Final Prototype	40
Conclusion	10
Appendices	10